

NAME: Ganesh Balkrushna Soanawane

ROLL NO:43

PROBLEM STATEMENT: In a company, employee salaries are stored in a list as floating-point numbers. Write a

Python program that sorts the employee salaries in ascending order using the following two algorithms:

- Selection Sort: Sort the salaries using the selection sort algorithm.
- Bubble Sort: Sort the salaries using the bubble sort algorithm.

After sorting the salaries, the program should display top five highest salaries in the company

'''

```
Salary=[]
```

```
size=int(input("Enter the array size:"));
```

```
for i in range(0,size):
```

```
    ele=float(input("Enter the element:"));
```

```
    Salary.append(ele)
```

```
def bubble_sort(Salary):
```

```
    for i in range(len(Salary)-1):
```

```
        for j in range(len(Salary)-i-1):
```

```
            if(Salary[j]>Salary[j+1]):
```

```
                temp=Salary[j];
```

```
                Salary[j]=Salary[j+1];
```

```
                Salary[j+1]=temp;
```

```
    return Salary;
```

```
def selection_sort(Salary):
```

```
    for i in range(0,size-1):
```

```
        for j in range(i+1,size):
```

```
            if(Salary[i]>Salary[j]):
```

```
                temp=Salary[i];
```

```
                Salary[i]=Salary[j];
```

```
                Salary[j]=temp;
```

```
    return Salary;
```

```
def menu():
```

```
    print("1.bubble sort:")
```

```
    print("2.selection sort:")
```

```
    print("3.Exit:")
```

```
while True:
```

```
    menu()
```

```
    choice=input("Enter your choice(1-3):")
```

```
    if choice=='1':
```

```
        sorted_bubblesort=bubble_sort(Salary)
```

```
        print("sorted_bubble sort is",sorted_bubblesort)
```

```
        len_b=len(sorted_bubblesort)
```

```
        print("The length of bubble sort is:",len_b)
```

```
        print("Sorted Salaries:",sorted_bubblesort)
```

```
        print("Top 5 salaries are:"sorted_bubblesort[-1:-6:-1])
```

```
    elif choice=='2':
```

```
        Sorted_SelectionSort=selection_sort(Salary)
```

```
        print("Sorted Selection sort is"Sorted_SelectionSort)
```

```
len_b=len(Sorted_SelectionSort)
print("The length of selection sort is:",len_b)
print("Top 5 salaries are:"Sorted_Selectionsort[-1:-6:-1])
elif choice=='3':
    print("Exit")
    break;
else:
    print("Invalid Choice")
```

'''Output:

```
gescoe@gescoe-OptiPlex-3020:~/SE divB $ python3 selection sort.py
```

Enter the array size:6

Enter the element:100

Enter the element:200

Enter the element:300

Enter the element:400

Enter the element:500

Enter the element:600

1.bubble sort:

2.selection sort:

3.Exit:

Enter your choice(1-3):1

sorted_bubble sort is [100.0, 200.0, 300.0, 400.0, 500.0, 600.0]

The length of bubble sort is: 6

Sorted Salaries: [100.0, 200.0, 300.0, 400.0, 500.0, 600.0]

Top 5 salaries are: [600.0, 500.0, 400.0, 300.0, 200.0]

1.bubble sort:

2.selection sort:

3.Exit:

Enter your choice(1-3):2

Selection sort not implement yet.

1.bubble sort:

2.selection sort:

3.Exit:

Enter your choice(1-3):3

Exit

'''